

CONDITIONAL PROBABILITY: LEARNING FROM NEW INFORMATION

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UPDATE PROBABILITIES USING NEW INFORMATION

In many real world situations, probabilities adjust as new, and most likely incomplete, information becomes available. Conditional probability provides a way to measure the likelihood of an event given that another event has already occurred, allowing us to incorporate additional information into our analysis.

Building on this idea, Bayes' formula provides a method for updating probabilities in light of this new evidence. These concepts are fundamental to fields such as statistics, machine learning, and quantitative finance, where decision-making often depends on continually reassessing uncertainty as new information arrives. In this article, we introduce the key definitions of conditional probability, Bayes' formula, and explore their practical applications.

BASIC DEFINITIONS AND THEOREMS

Let us start with the basics. **Conditional probability** is denoted by $P(A | B)$, representing the probability of event A occurring given that event B has already occurred. If $P(B) > 0$, then $P(A | B) = \frac{P(A \cap B)}{P(B)}$. In other words, $P(A | B)$ is the proportion of outcomes in B that also belong to A . Rearranging the definition of conditional probability gives $P(A \cap B) = P(A | B)P(B)$. This result is known as the **multiplication rule**. It states that the probability of both events A and B occurring is equal to the probability that B occurs, multiplied by the probability that A occurs given that B has already occurred. More generally, with events $A_1, A_2, \dots, A_n$, then $P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2 | A_1)P(A_3 | A_1 \cap A_2) \dots P(A_n | A_1 \cap \dots \cap A_{n-1})$.

In many probability problems, an event can occur through several different routes. The **Law of Total Probability** allows us to calculate the probability of an event by considering all possible ways in which it can occur. Let $B_1, B_2, \dots, B_n$ be mutually exclusive events that form a partition of the sample space, that is $B_i \cap B_j = \emptyset, \forall i \neq j$; $\bigcup_{i=1}^n B_i = \Omega$. Then, for any event A , $P(A) = P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_n) = \sum_{i=1}^n P(A | B_i)P(B_i) = P(A | B_1)P(B_1) + P(A | B_2)P(B_2) + \dots + P(A | B_n)P(B_n)$. This result follows directly from the multiplication rule. Since the events $B_1, B_2, \dots, B_n$ cover the entire sample space, the event A can be decomposed as $A = (A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_n)$. The probability of A is therefore obtained by summing the probabilities of these mutually exclusive events, giving the Law of Total Probability.

Two events A and B are said to be independent if the occurrence of one event does not affect the probability of the other. Mathematically, A and B are independent if $P(A | B) = P(A)$. Equivalently, using the multiplication rule, $P(A \cap B) = P(A)P(B)$. Independence is an important concept in probability theory, as it often simplifies calculations and forms the basis of many probabilistic models.

One of the most important results in probability theory is **Bayes' Formula**, which allows us to update probabilities when new information becomes available. It follows directly from the definition of conditional probability and the multiplication rule. We obtain $P(A | B) = \frac{P(B|A)P(A)}{P(B)}$, provided that $P(B) > 0$. This result is known as Bayes' Formula. Bayes' Formula enables us to revise the probability of an event A after observing evidence B . In this context, $P(A)$ is known as the prior probability, $P(B | A)$ as the likelihood, and $P(A | B)$ as the posterior probability. More generally, we can write $P(B_j | A) = \frac{P(A|B_j)P(B_j)}{\sum_{i=1}^n P(A|B_i)P(B_i)}$ if $B_i, i = 1, \dots, n$ are mutually

exclusive events whose union is the entire sample space, Ω . Note that not all conditional probability problems have intuitive solutions, many demand logical analysis instead.

AN EXAMPLE: MEDICAL TESTING

A famous example is the medical test example, this is because it shows how intuition can sometimes be misleading.

Suppose that 1% of a population has a particular disease. A test for the disease is 99% accurate, meaning: $P(\text{Positive} | \text{Disease}) = 0.99$, and $P(\text{Positive} | \text{No Disease}) = 0.01$. If a person tests positive, what is the probability that they actually have the disease?

Solution: Let D = the person has the disease, and $+$ = the test result is positive. Using Bayes' Formula, $P(D | +) = \frac{P(+|D)P(D)}{P(+)}$. The denominator is found using the Law of Total Probability, $P(+) = P(+ | D)P(D) + P(+ | D^c)P(D^c)$. Substituting the values, we obtain $P(+) = (0.99)(0.01) + (0.01)(0.99) = 0.0198$. Therefore, $P(D | +) = \frac{(0.99)(0.01)}{0.0198} = 0.5$. So even after receiving a positive test result, the probability of actually having

the disease is only 50%. This surprising result highlights the importance of accounting for the underlying prevalence of an event and demonstrates the power of Bayes' Formula in updating probabilities when new information becomes available.

FROM PROBABILITIES TO PREDICTIONS

Rather than relying solely on historical data or initial assumptions, we can continuously revise our estimates in response to new evidence. This idea is particularly important in quantitative finance. Investors and traders regularly receive new information in the form of earnings reports, economic data releases, and market signals. Bayes' Formula provides a way to incorporate this information into probabilistic models, allowing predictions to evolve as market conditions change. More broadly, this process of updating probabilities forms the foundation of Bayesian statistics, machine learning, and modern decision-making under uncertainty.

CONCLUSION

Conditional probability allows us to determine the likelihood of an event given additional information, providing a foundation for reasoning under uncertainty. By developing the multiplication rule, the Law of Total Probability, and Bayes' Formula, we have seen how probabilities can be updated as new evidence becomes available. These concepts are central to probability theory and have applications across various fields, where the ability to revise beliefs and make informed predictions is essential. Understanding conditional probability and Bayes' Formula therefore provides a crucial step towards more advanced probabilistic and statistical modelling.

ALEXANDER FLANN